

Introductory Course on Automation and its Use for Interactive Systems Design and Engineering

Philippe Palanque

ICS-IRIT, University Toulouse 3, 118, route de Narbonne, 31062 Toulouse, France

ABSTRACT

This course takes both a practical and theoretical approach to introduce the principles, methods and tools for automation. Examples are taken from industries that have been embedding automation in their systems for a long time (such as aviation, automotive, satellite ground segments . . .), but the main focus is on designing and assessing automation for interactive systems. Interactive hands-on exercise of how to "do it right", will be provided answering questions such as: How to add automation in interaction techniques to improve efficiency? How to migrate tasks towards automation to improve effectiveness? How to design usable automation at system, application and interaction levels? Is more automation always better? If not, when do we stop automating? And more....

CCS CONCEPTS

• D.2.2 [Software] Design Tools and Techniques; • H.5.m. Information interfaces and presentation (e.g., HCI);

KEYWORDS

Automation, Interactive systems engineering, Interactive systems design, Usability and Reliability of partly-autonomous systems

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1 INTRODUCTION

Early approaches in automation were focusing on allocating basic functions to the best player (e.g. Fitts' approach Machine Are Better At – Men Are Better At) [5] and more than 50 years later these findings were still used and transposed to computing systems [15]. Such concept of allocation of function to best actor was however deeply criticized in [16] as it misses (among others) the perspective of teamwork between automation and the operator. This course builds on top of this perspective adopting the human-in-the-loop perspective and considering automation as an artefact supporting operators' work and users' tasks and not a goal per se with the objective or getting rid of the users that are a source or errors as detailed in [19].

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This course thus focuses on the use of the concept of automation for interaction design and for interactive systems designs. It advocates the point of view of automation as a design alternative resulting on the migration of actions previously performed by users to actions performed by a system. Current push in automation is towards fully autonomous systems (such as Tesla cars) raising critical issues such as: how to make it possible to users to foresee future states of the automation, how to disengage automation or how to make sure that users are able to take over when automation fails Beyond, introducing automation in users' work and activities changes deeply its own nature getting closer and closer to supervision, which is a different interaction and work paradigm.

Supervision activities are very different from directly performing activities and the migration of activities to the system might have severe consequences:

- in terms of overall performance of the couple (user, system),
- in terms of skilling (and especially deskilling) [9],
- in terms of errors related to the very nature of human behavior [8], in terms of training [4] . . .

This course promotes automation as a mean and not automation as a goal and takes a practical approach to introduce attendees to the principles, methods and tools for the design and the assessment of automation within the design of interactive systems.

2 CONTRIBUTION AND BENEFIT

This course intends to provide newcomers with background knowledge in automation. It provides an overview on how the recent advances in automation can be exploited to design and assess interactive systems. This course will also provide material for lecturers who want to include automation in their HCI courses. Finally, the course will provide examples from various domains including aviation, Air Traffic Management, Automotive highlighting both successes and failures of automation. Concrete examples and exercises will also be provided to support the application of the course elements to practical cases.

3 OBJECTIVES

On completion of this tutorial, attendees will:

- Know the intrinsic and extrinsic characteristics of automation, [2, 5, 10]
- Be able to use AFAR paradigm (Allocation of Functions, Authority and Responsibility) [13] when designing and assessing automation for interactive systems,
- Know many examples on how automation has been used in various domains such as automotive, aviation, Air Traffic Control, space domain . . . [11]

- Know and understand that automation is already pervasive in interactive systems and that the consequences are usually underestimated, [2]
- Have experience in analyzing an interactive systems focusing on the tasks users have to perform with it and how automation impacts user's work and activities,
- Know the main objectives of automation and the design rules that must be followed to design usable partly-autonomous interactive systems [18]
- Know the danger and the benefits of more automation in interactive systems [12].

4 DESCRIPTION AND CONTENT

This course focuses on the foundations of automation and how this concept can be used while designing and assessing interactive systems

- Introduction to automation, its basic principles and how this concept can be decomposed in components [10]. Beyond, we will present the analysis of multiple CHI papers that proposed new interaction techniques adding automation (usually without acknowledging the proposal as steps towards more autonomous interactions).
- Basic principles for describing user activity and tasks (hierarchical view on human activities, abstraction and refinement, temporal ordering, objects, information and knowledge . . . [6] and also about human activity with a focus on automation [14],
- Overview of automation designs and how they are pervasive in interactive systems and interaction techniques as well as their desired properties (intelligibility, transparency [2], continuity [20] . . .)
- Overview of automation principles that are used and deployed in various domains as well as their intended objectives (enhance user, protect the user via safety or security nets, increase user comfort, entertain used, protect against automation . . .).
- List of design principles for the identification of users' activities that could be good candidates for task migration towards automation, authority sharing, impact of automation degradation on tasks performance)
- Taking into account human errors [1] at design time as well as possible degradation of automation [3] to assess relevance of automation in a given context (identification of types, location and likelihood of human errors)

5 AGENDA

This course is composed of one single course unit going from conceptual to very practical. It includes interactive hands-on exercises, case studies and practice on automation design and assessment in the content of interactive systems design.

6 AUDIENCE AND PREREQUISITE

This course is open to researchers, practitioners, educators and students of all experience levels. No specific skills or knowledge are required beyond a background in User Centered Design. Paper [14] should be read before attending the lecture.

7 COURSE BACKGROUND

Two closely related courses have been given at CHI, but content and philosophy of this one is very different. This third edition takes into account the feedback from participants of the previous editions. The first course was about "Designing and Assessing Interactive Systems Using Task Models" (it was given at CHI 2015 and was using automation as a design alternative for tasks allocation) [17]. Feedback from participants showed that they would have a strong interest in automation per se (both foundations and applications) with a focus on autonomous cars. A second course was proposed at CHI 2018 about the risks and benefits of higher levels of automation and was entitled "Automation: Danger or Opportunity? Designing and Assessing Automation for Interactive Systems" [12]. Again, feedback from attendees showed another interest in the use of automation in interaction design and in user interface design which was only touched by some examples of successes and failures. This third course focuses on these specific aspects building from failures and successes in automation design and engineering in safety critical contexts. As an example a longer version of this course it taught at Eurocontrol¹.

8 INSTRUCTOR'S BACKGROUND

The instructor has been involved in the design and assessment of automation in multiple projects (mostly in safety critical systems including Air Traffic Management, Satellite ground segments and cockpits of large civil aircrafts).

Philippe Palanque Philippe Palanque is professor in Computer Science at the University Toulouse 3 and is head of the ICS (Interactive Critical Systems) research group at IRIT Lab. Since the early 90's his research focus is on interactive systems engineering proposing notations, methods and tools to integrate multiple properties such as usability, dependability, resilience and more recently user experience. These contributions have been developed together with industrial partners from various application domains such as civil aviation, air traffic management or satellite ground segments. Recently he has been involved in the specification of future interactive cockpits and their interactions and in the modelling of operational states of civil aircraft (with direct support from and close collaboration with Airbus). The main driver of Philippe's research over the last 20 years has been to address in an even way Usability, Safety and Dependability in order to build trustable safety critical interactive systems. He has been working in the area of automation for more than ten years, was a member of the SESAR Higher Automation Levels in Aviation network of excellence and paper co-chair of ATACCS (Application and Theory of Automation in Command and Control Systems) 2015 conference. He was steering committee chair of the CHI conference series at ACM SIGCHI, is a member of the CHI academy and chair of IFIP Technical Committee on Human-Computer Interaction (TC13). He edited and co-edited more than twenty books or conference proceedings including the "Handbook on Formal Methods in Human-Computer Interaction" published by Springer in 2017.

¹<https://trainingzone.eurocontrol.int/ilp/pages/coursedescription.jsf?catalogId=896431&courseId=6629160>

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